SIADH in the Elderly: A Complication of Pneumonia

Lakeland Regional Health
GRADUATE MEDICAL EDUCATION

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BACKGROUND

- Community-acquired pneumonia (CAP) is the most frequent infectious cause of hospitalization and mortality in industrialized countries.
- The incidence of hospitalization due to CAP is about 3 cases per 1,000 adults and those that are hospitalized have a mortality rate around 10%.
- Hyponatremia is the most common electrolyte imbalance seen in clinical practice.
- Studies have shown that on average, 4% of patients presenting with pneumonia will have hyponatremia due to SIADH.
- Hyponatremia in patient with pneumonia has been proven to be a poor prognostic indicator – associated with increased hospital mortality, prolonged hospital stays and increased rates of readmission.
- Geriatric patients are at the highest risk for developing hyponatremia.
- Emerging evidence suggests that the presence of hyponatremia in pneumonia may serve as a marker for systemic inflammation and cytokine dysregulation.
- Correcting hyponatremia in pneumonia patients has been shown to reduce the risk of severe complications, but aggressive correction can also pose risks, particularly in the elderly.

CASE PRESENTATION

A female in her 90s with a past medical history of hypertension, hyperlipidemia, bladder prolapse, and glaucoma presented to the emergency department with a productive cough and generalized weakness for two weeks. She also endorsed a decreased appetite, decreased water intake, and urinary hesitancy. Patient did not have any neurological deficits or change in mentation. On admission, her blood pressure was elevated at 172/83. Physical exam showed decreased breath sounds and conversational dyspnea as well as poor skin turgor and dry mucous membranes. Pertinent labs were leukocytosis of 13.92, sodium of 122, lactic acid of 2.7, and BNP of 1399. Chest X-Ray showed mild right basilar infiltrate and minimal left basilar atelectasis. CT scan of chest showed dense consolidation of both lower lobes of the lungs which had air bronchograms as well as thickening of bronchial walls. Further testing revealed an elevated urine sodium of 69, serum uric acid of 2.1, and urine and serum osmolality of 343 mOsm/kg and 256 mOsm, respectively. Based on the lab results, a diagnosis of SIADH most likely secondary to pneumonia was made. Initial management included fluid restriction to 1.5 liters and administration of 1 gram of salt tablets three times daily, but these measures resulted in minimal improvement in serum sodium levels. On the third day of hospitalization, a single dose of tolvaptan was administered, which effectively raised her sodium levels from 124 mEq/L to 132 mEq/L. The patient's clinical condition improved, and she was discharged with a one-day course of azithromycin for pneumonia. Importantly, she was advised to follow up with repeat imaging to ensure that the SIADH was not secondary to an underlying lung mass.

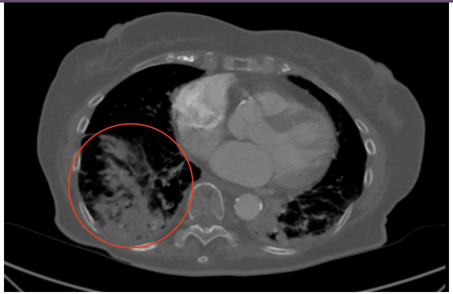

Figure 1: CT chest without contrast demonstrates moderate amount of dense consolidation and air bronchograms in right lower lobe (both circled in red), as well as in the left lower lobe.

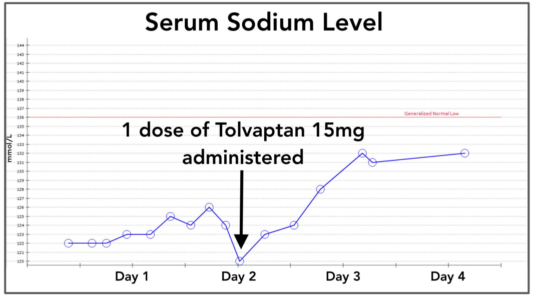

Figure 2: Serum Na level increases sharply after administration of a single 15mg dose of Tolvaptan.

CONCLUSION

This case highlights the complex interplay between CAP and SIADH in a geriatric patient. Pneumonia can serve as a significant trigger for SIADH, leading to hyponatremia—a condition that is not only common but also associated with worse clinical outcomes, particularly in the elderly. Prompt recognition and appropriate management can prevent serious complications. Fluid restriction remains the first-line treatment, but as illustrated, it may not always be sufficient, necessitating the use of pharmacologic agents such as tolvaptan. However, such interventions require careful monitoring to avoid rapid correction and its associated risks. Moreover, this case emphasizes the need for comprehensive follow-up care, including repeat imaging, to ensure that the SIADH is not secondary to an underlying malignancy or other pulmonary pathology that could have been masked by the pneumonia.

DISCUSSION

- The pathophysiology as to how pneumonia can cause SIADH is still being studied. However, several mechanisms have been proposed:
- Pneumonia causes lung injury, which leads to pulmonary vasoconstriction thus reducing left atrial filling causing decreased atrial receptor firing signaling the hypothalamus to increase ADH.
- Inflammatory cytokines such as Interleukin-6 (IL-6) and Interleukin-1b (IL-1b) have been shown to stimulate non-osmotic secretion of ADH as well damage the alveolar basement membrane, leading to pulmonary vasoconstriction, thus increasing ADH via the mechanism discussed above. While this mechanism has been extensively studied in COVID-19 pneumonia, studies are showing that it may play a significant role in CAP as well.
- Pressure on the diaphragm during pneumonia can trigger the release of ADH.
- Fluid restriction with salt tabs was implemented as the first-line therapy, however, minimal response in serum sodium was seen, thus, tolvaptan was considered to safely correct sodium levels.
- Tolvaptan should be used in patients when fluid restriction alone is insufficient or when rapid correction is necessary due to symptomatic hyponatremia (e.g. seizures, altered mental status).
- The use of tolvaptan requires careful monitoring to avoid overly rapid correction of sodium, which carries its own risks, particularly osmotic demyelination syndrome (ODS), therefore sodium should not be corrected more than 8-12 mEq/L in the first 24 hours.
- The primary treatment should focus on managing the pneumonia, which is essential in resolving SIADH as studies have shown that appropriate antibiotic treatment of the pneumonia alone can resolve the SIADH.
- Patients can be discharged once serum sodium levels have stabilized and symptoms have resolved; one dose of tolvaptan is sufficient in some cases, but continued monitoring and follow-up are crucial to ensure long-term stability and prevent recurrence of hyponatremia.
- It is also important to note that sometimes imaging cannot detect an underlying lung mass due to the presence of overlying pneumonia. Therefore, patients should undergo follow-up imaging after discharge, once the pneumonia has resolved, to rule out any potential underlying lung pathology.

REFERENCES

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